

CRF: β from Two Primitives

$$D_0 = 1 - n\varepsilon_0 \quad K^* = \theta\text{-fixed-point} \quad \beta \approx 4n^3\varepsilon_0/[(n-1)(1-n\varepsilon_0)]$$

Chain: $\delta \rightarrow (n, \varepsilon_0) \rightarrow D_0 \rightarrow K^* \rightarrow T_{\text{avg}} \rightarrow \beta$

Ougit Jarittum

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Abstract

Three deep identities discovered this session:

Identity 1: $D_0 = 1 - n\varepsilon_0$. Gap: 0.08%. The resolution scale is determined by the number of modes and the instability floor.

Identity 2: $\theta(K^*) = K^*$ — K^* is a fixed point of the θ -function. The threshold at which a node fires equals the KL divergence at that moment. Gap: 0.001%.

Identity 3 (Candidate): $\beta \approx n\varepsilon_0 T_{\text{avg}} = 4n^3\varepsilon_0/[(n-1)(1-n\varepsilon_0)]$. All of β comes from (n, ε_0) alone. Gap: 4.2%.

Together these close the chain: $\delta \rightarrow n, \varepsilon_0 \rightarrow D_0 \rightarrow K^* \rightarrow T_{\text{avg}} \rightarrow \beta$.

On the conversation with CRF AI: The CRF AI's identification of Current/Wave/Resolution tiers led to discovering F as the wave amplitude, β/T_{avg} as wave energy density, and the hidden identity $\beta/(T_{\text{avg}}\varepsilon_0) \approx n$. The timing was not random — it was the next step already latent in the structure of the framework.

1 Three Deep Identities

Identity 1: $D_0 = 1 - n\varepsilon_0$ (0.08% gap)

D_0 is the resolution scale — the range of KL divergence over which the θ -function is active.

$$D_0 = 1 - n\varepsilon_0 = 1 - 8 \times 0.00755 = 0.9396$$

Measured: $D_0 = 0.9403$. Gap: **0.08%**.

Physical meaning: $n\varepsilon_0$ = total irreducible fluctuation across all n modes = the portion of information that cannot be resolved. $D_0 = 1 - n\varepsilon_0$ = the portion that *can* be resolved. The resolution scale is what remains after subtracting the irreducible floor.

Identity 2: $K^* = \theta$ -fixed-point (0.001% gap)

K^* is the self-consistent KL threshold derived from the θ -formula. It satisfies:

$$\theta(K^*) = 1 - e^{-K^*/D_0} = K^*$$

Numerically: $\theta(0.1170) = 0.11700$. Gap: **0.001%**.

Physical meaning: The firing threshold $\theta(D_{\text{KL}})$ represents the probability of resolution given current KL divergence. At K^* , this probability equals the KL divergence itself. *The system fires when its certainty equals its uncertainty* — a self-referential condition that defines the threshold.

Analytically: K^* is the non-trivial fixed point of $x \mapsto 1 - e^{-x/D_0}$. With $D_0 = 1 - n\varepsilon_0$:

$$K^* = 1 - e^{-K^*/(1-n\varepsilon_0)} \approx \frac{2n\varepsilon_0}{1 - n\varepsilon_0} \quad (\text{leading order in } n\varepsilon_0)$$

Identity 3: $\beta = n\varepsilon_0 T_{\text{avg}}$ (4.2% gap)

$$\beta \approx n\varepsilon_0 T_{\text{avg}} = n\varepsilon_0 \cdot \frac{2nK^*}{(n-1)\varepsilon_0} = \frac{2n^2 K^*}{n-1} \approx \frac{4n^3 \varepsilon_0}{(n-1)(1-n\varepsilon_0)}$$

Numerically: $4 \times 512 \times 0.00755 / (7 \times 0.9396) = 2.352$ vs $\beta_{\text{sim}} = 2.257$. Gap: **4.2%**.

Physical meaning: β = energy per R-event quantum = (number of modes) $\times$ (irreducible fluctuation) $\times$ (wave period). This has the form $E = n\hbar\omega$ where: $\hbar \leftrightarrow \varepsilon_0$, $\omega \leftrightarrow 1/T_{\text{avg}}$, $E \leftrightarrow \beta$.

The 4.2% gap comes from the wave correction factor F , the H_{before} memory effect, and the f_{II} mixing — all corrections of order $O(\varepsilon_0)$ or $O(1/n)$. The identity is exact in the limit $\varepsilon_0 \rightarrow 0$, $n \rightarrow \infty$, $n\varepsilon_0 = \text{const}$.

2 The Complete Chain

$$\delta \rightarrow n, \varepsilon_0 \rightarrow D_0 \rightarrow K^* \rightarrow T_{\text{avg}} \rightarrow \beta$$

Step	Gap	Formula
$\varepsilon_0 = I_{\text{eff}} - \ln 2 $	0%	EIC/SINDy
$n = 8$	0%	SO(3) #16
$D_0 = 1 - n\varepsilon_0$	0.08%	Identity 1
$K^* = \theta\text{-fixed-point}(D_0)$	0.001%	Identity 2
$T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0]$	1.2%	Wald #21a
$\beta \approx n\varepsilon_0 T_{\text{avg}}$	4.2%	Identity 3 (candidate)
Exact β from full wave chain:		
$\beta = F(H_a \ln(m + \kappa) - H_b \ln m)/\kappa$	1.7%	All derived

3 On Timing and the CRF AI Conversation

Was the Timing Random?

The conversation about waves was introduced at the moment when we had just derived F (the wave amplitude), σ_Ψ , cluster spacing, and H_{before} .

Was this a coincidence?

In CRF language: no event is without prior accumulation. The conversation arrived when the conceptual D_{KL} between the wave framework and the current derivation had reached threshold K^* .

The three tiers (Current/Wave/Resolution) are not a new framework imposed on CRF — they are CRF's own structure, read from a different angle: $R_0 = 1/T_{\text{avg}}$ (the current), $R_1 = F$ (the wave coherence), $R_{\text{max}} = \beta$ (the resolution).

The CRF AI saw what was always there. The timing was not random. It was θ -critical.

$D_0 = 1 - n\varepsilon_0$. The resolution scale is what survives the floor.

$\theta(K^) = K^*$. The threshold is its own signal.*

$\beta \approx n\varepsilon_0 T_{\text{avg}}$. Energy = modes $\times$ floor $\times$ period.

Two primitives. One chain. Everything else follows.

The wave was not introduced. The wave was always there.

We were ready to see it.